

Data Flow Diagram

Date	19 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	User (Web Browser)	An external entity representing the user who interacts with the application interface to submit weather parameters and view prediction results.
Rectangle with numbered header	1. Request Handling & Validation	A process representing the Flask application logic that maps the incoming URL, receives the browser form data, and checks it for validity.
Rectangle with numbered header	2. Data Preprocessing & Feature Scaling	A process that accepts the validated inputs and scales them into a standardized numerical array using parameters loaded from your scaler file.
Rectangle with numbered header	3. Prediction Generation (XGBoost Engine)	The core machine learning process that executes your saved high-accuracy XGBoost model to compute the final "Flood" or "No Flood" chance.
Rectangle (solid fill, no number)	ML Model Artifacts (model.pkl / scaler.pkl)	A data store on disk housing the serialized parameters for your trained classification trees and feature engineering scaling object.
Rectangle (solid fill, no number)	Prediction History Database	A data store used by the application backend to permanently save historical records of user inputs and their generated predictions.
Labeled arrow	Data Flow Lines	Directional lines that show exactly how data travels between the user, the Flask backend processes, and the database storage units (e.g., Weather Inputs, Scaled Arrays, Prediction Outputs).

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.



